

Fairness in shallow neural network models for crime prediction and location

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Abstract—This project proposes the development of a machine learning model based on shallow neural networks, utilizing a Radial Basis Function (RBF) network and Convolutional Long Short Term Memory (ConvLSTM) to localize criminal events in the city of Chicago. The analyzed data spans historical records from 2001 to 2024, with a specific focus on recent years, and targets three high-incidence crime categories: Theft, Battery, and Criminal Damage. These data are processed into 1x1 km spatial grids to capture detailed spatiotemporal patterns.

The project also prioritizes fairness in predictions by evaluating error distribution between high- and low-income populations, classified based on average household income. This ethical analysis aims to identify potential biases in predictions, contributing to the discourse on the application of shallow neural networks in sensitive contexts. Using a sliding window methodology and weekly evaluations—aligned with standard practices employed by law enforcement agencies—the model’s accuracy is assessed to provide evidence supporting the development of more equitable and effective public policies for crime management.

I. PROJECT GOAL AND SCOPE

The objective of this project is to develop a machine learning model based on shallow neural networks, specifically utilizing a Radial Basis Function (RBF) and Convolutional Long Short Term Memory (ConvLSTM) network. These networks will address two key challenges in crime management in the city of Chicago: predicting crime intensity and spatially localizing criminal events. The project focuses on exploring the capability of shallow neural networks to model spatiotemporal patterns in criminal activities and investigating how these architectures may be susceptible or resilient to bias in error distribution across advantaged and disadvantaged populations.

The analysis will leverage historical data spanning from 2001 to the present, with a primary focus on the years 2022-2024 to mitigate potential seasonal effects in crime patterns. Three high-incidence and socially relevant crime categories have been selected: Theft, Battery, and Criminal Damage. The data will be processed into 1x1 km spatial grids, providing sufficient granularity to analyze local and regional patterns. Additionally, spatial fairness will be evaluated by analyzing the average prediction error across different populations and its relationship to socioeconomic and demographic factors.

The ethical focus of this project aims to ensure that the model does not perpetuate historical inequalities by analyzing how model-driven decisions may disproportionately impact disadvantaged communities. This effort seeks to contribute to the development of fairer tools for crime prediction and

management, with the potential to influence more equitable public policy-making.

II. LITERATURE REVIEW

A. Background on Crime Prediction Models

The application of machine learning to crime prediction has evolved significantly over the past decade. Traditional hotspot mapping approaches have given way to sophisticated neural network-based methods capable of capturing complex spatiotemporal patterns in criminal activity [1], [2]. These models have demonstrated impressive accuracy in predicting various types of crimes, from property theft to violent incidents. A notable example is the implementation in Los Angeles, where predictive policing systems achieved a 7.4% reduction in crime rates during their first year of deployment [3].

Recent advances in deep learning have enabled the development of models that can incorporate multiple data streams, including historical crime data, demographic information, weather patterns, and urban infrastructure details [4], [5]. The integration of diverse data sources has significantly improved prediction accuracy, as demonstrated by systems like CrimeScan in Chicago, which combines over 20 different data sources [6]. However, these advancements have also raised concerns about the potential perpetuation and amplification of existing societal biases through automated prediction systems.

B. Fairness in Machine Learning

The concept of fairness in machine learning has emerged as a critical consideration, particularly in applications that directly impact human lives and social systems [7]. Research has revealed multiple dimensions of fairness, including individual fairness, where similar individuals should receive similar predictions; group fairness, ensuring equivalent treatment across demographic groups; temporal fairness, maintaining consistent fairness over time; and spatial fairness, providing unbiased attention across geographic locations [8].

A comprehensive study by Richardson et al. [9] demonstrated how prediction systems can perpetuate historical biases, finding false positive rates up to three times higher in predominantly minority neighborhoods. This disparity highlights the complex interaction between algorithmic decision-making and existing social inequities, particularly in urban environments with historical patterns of segregation and inequality [10].

C. Geographical and Socioeconomic Factors

Urban crime patterns exhibit strong correlations with socioeconomic factors, creating significant challenges for developing unbiased prediction models [11]. The relationship between poverty rates and crime reporting, variations in police presence and response times across neighborhoods, and the impact of infrastructure and urban planning on crime patterns all contribute to the complexity of fair prediction systems.

Recent research has highlighted the particular importance of boundary areas between different socioeconomic regions. A longitudinal study in Baltimore revealed that areas at the boundaries between neighborhoods of different socioeconomic levels experienced crime rates up to 23% higher than the city average [12]. This finding emphasizes the critical nature of understanding and addressing fairness issues in these transition zones.

III. CASE STUDY

The groundbreaking work by Ensign et al. [12] serves as a foundational case study for our research. Their investigation of runaway feedback loops in predictive policing systems revealed critical insights into how these systems can perpetuate and amplify existing biases. Through detailed analysis of real-world deployment data, they demonstrated conclusive evidence of self-fulfilling predictions in areas of increased police presence. Their work also showed how initial biases can be amplified over time through feedback loops, with particularly pronounced effects at the boundaries between different policing zones.

The study's findings regarding differential impacts on various demographic groups provide crucial insights for our research, especially in understanding how predictive models influence real-world policing decisions and their subsequent effects on different communities. Their methodology for identifying and measuring boundary effects between different policing zones offers a valuable framework for our investigation of fairness at socioeconomic boundaries.

IV. TRAINING DATA SET

Our research will leverage the comprehensive Chicago Data Portal's crime dataset [13], providing detailed crime records spanning from 2001 to the present. This dataset offers extensive temporal coverage with over 20 years of historical data, precise spatial granularity through GPS coordinates for each incident, and a rich feature set including crime type, description, location type, and timing information, along with comprehensive community context through beat and district information.

To enhance the primary dataset's utility, we will integrate additional data sources including socioeconomic indicators from the American Community Survey, census tract-level demographic data, local infrastructure and urban planning information, and historical weather data. Our data processing pipeline will implement robust handling of missing values and outliers, multi-scale temporal aggregation, sophisticated

spatial clustering and boundary detection, and careful feature engineering for socioeconomic indicators.

V. DATA PRE-PROCESSING AND EXPLORATORY DATA ANALYSIS

A. Data Cleaning and Pre-processing

The dataset covers crime reports from 2001 to 2024 in Chicago, with a primary focus on the years 2022-2024. The following actions were performed during data pre-processing:

- **Duplicate Removal:** Duplicate records were excluded to prevent data redundancy.
- **Handling Missing Values:** Records with missing critical information (e.g., *Primary Type*, *Location Description*) were either removed or imputed.
- **Categorical Encoding:** Variables such as *Primary Type* and *Location Description* were numerically encoded.
- **Temporal Segmentation:** Data was organized into quarterly windows to enable temporal analysis and improve model training.

B. Feature Selection and Engineering

To enhance model performance, features were carefully selected and engineered:

- **Continuous Features:** Variables such as *Latitude*, *Longitude*, and *Date* were scaled to a range of [0, 1].
- **Categorical Features:** One-hot encoding was applied to features like *Primary Type* and *Location Description*.
- **Derived Features:** New features such as *Crime Rate per Km²* and *Weekend Indicator* were created to capture temporal and spatial patterns.

C. Exploratory Data Analysis

1) *Crime Categories:* An analysis of crime reports (2001-2024) highlighted three main categories: *Theft*, *Battery*, and *Criminal Damage*, which together accounted for over 50% of all reported crimes. These categories were prioritized for modeling.

TABLE I
CRIME CATEGORIES ANALYSIS (2001-2024)

Crime Type	Occurrences	Percentage
Theft	311,148	31.76%
Battery	247,818	25.31%
Criminal Damage	143,245	14.63%
Assault	96,127	9.82%
Deceptive Practice	94,718	9.67%
Other Offenses	86,217	8.81%
Total	979,273	100%

2) *Crime Locations:* The analysis revealed that most crimes occurred on streets, in residences, and in apartments, emphasizing the importance of these locations for model inputs.

3) *Temporal Patterns:* Crimes showed seasonal variations, with notable peaks during summer months and weekends. These temporal patterns suggest the importance of incorporating time-sensitive features into the model.

TABLE II
CRIME LOCATIONS ANALYSIS (2001-2024)

Location Description	Occurrences	Percentage
Street	297,259	28.14%
Residence	223,296	21.13%
Apartment	173,419	16.41%
Sidewalk	113,895	10.78%
Other	54,601	5.16%
All Others Combined	192,803	18.38%
Total	1,055,273	100%

D. Statistical Analysis and Visualization

- **Feature Distribution:** Histograms and box plots revealed that variables like *Crime Density* followed a right-skewed distribution.
- **Correlation Analysis:** Heatmaps demonstrated strong correlations between population density and theft occurrences.
- **Geospatial Visualization:** Heatmaps showed higher crime concentrations in low-income districts, highlighting potential fairness concerns.

E. Key Observations

- *Theft, Battery, and Criminal Damage* are the most frequent crime types and warrant special focus in predictive modeling.
- Crimes are concentrated in specific locations (e.g., streets, residences), suggesting spatial dependencies.
- Temporal trends indicate the need for dynamic models to account for seasonality and time-sensitive patterns.

F. Population Segmentation: Protected and Unprotected Populations

To analyze disparities in model performance, the populations (districts of Chicago) were segmented into *protected* and *unprotected* groups based on their median household income. Using the Chicago family income survey, the median income across all districts was determined to be \$21,323. Districts with a median income below this threshold were classified as **unprotected**, while those with a median income above this threshold were classified as **protected**. This classification reflects the socioeconomic advantages or disadvantages experienced by these populations.

a) *Protected Populations*:: Districts classified as protected (higher income) include:

- Douglas, Grand Boulevard, Kenwood, Lincoln Square, Hyde Park, Rogers Park, Jefferson Park, Forest Glen, North Park, Portage Park, Irving Park, Dunning, Montclare, West Ridge, Logan Square, West Town, Near West Side, Uptown, Near South Side, Norwood Park, Near North Side, Loop, Avalon Park, Lake View, Calumet Heights, North Center, Hegewisch, Garfield Ridge, Bridgeport, Clearing, Lincoln Park,

Ashburn, Beverly, Mount Greenwood, Morgan Park, O'Hare, Edgewater, Edison Park.

b) *Unprotected Populations*:: Districts classified as unprotected (lower income) include:

- Oakland, Fuller Park, Washington Park, Woodlawn, Albany Park, Belmont Cragin, Hermosa, Avondale, Humboldt Park, Austin, West Garfield Park, East Garfield Park, North Lawndale, South Lawndale, Lower West Side, Armour Square, South Shore, Chatham, South Chicago, Burnside, McKinley Park, Roseland, Pullman, South Deering, East Side, West Pullman, Riverdale, Archer Heights, Brighton Park, New City, West Elsdon, Gage Park, West Lawn, Chicago Lawn, West Englewood, Englewood, Greater Grand Crossing, Auburn Gresham, Washington Heights.

c) *Geographic Implications*:: This segmentation provides a framework to evaluate fairness in the model's predictions. Each district was further divided into 1km x 1km grid cells, inheriting the classification of its parent district. This fine-grained segmentation allows the model to identify disparities not only between districts but also within smaller geographic units. By linking predictive performance to income-based classifications, we aim to uncover systematic biases that may disadvantage unprotected populations, offering insights into the socioeconomic implications of predictive policing systems (fig. 1).

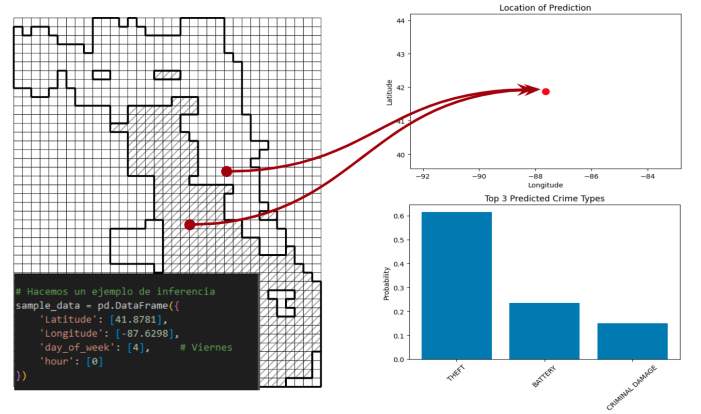


Fig. 1. Model prediction and segmentation example for Chicago.

VI. PROPOSED SHALLOW NEURAL NETWORK BASED APPROACH

A. Neural Network Architecture and Rationale

The proposed model shallow is combines a custom-designed Radial Basis Function (RBF) layer to address the specific requirements of crime prediction:

- **RBF Component:** The RBF layer is designed to enhance spatial feature representation by modeling local proximity relationships using a Gaussian activation function.

This architecture was chosen to capitalize on the RBF's proficiency in capturing spatial relationships.

The proposed model deep is ConvLSTM approach to address the specific requirements of crime prediction from spatiotemporal data:

- **ConvLSTM Component:** The ConvLSTM layers extend traditional LSTM units with convolutional operations over spatial dimensions, effectively capturing both the temporal progression of crime data and the localized spatial patterns in a unified framework.

This architecture was chosen to leverage the ConvLSTM's capacity to learn complex spatiotemporal features, thereby improving predictive performance in scenarios where spatial structure (e.g., city grids) and temporal dynamics (e.g., daily crime evolution) are intertwined.

B. Feature Engineering and Input Selection

The feature engineering process was designed to extract meaningful spatial, temporal, and categorical attributes:

- **Selected Features:**
 - *Latitude* and *Longitude* to capture spatial information.
 - *Day of the Week* and *Hour* to encode temporal dependencies.
- **Label Encoding:** The target variable (*Primary Type*) was encoded using a label encoder to convert crime types into numerical categories.
- **Normalization:** Features were scaled to a range of [0, 1] using Min-Max scaling to ensure stable and efficient training.

C. Network Architecture

1) *shallow NN-based:* The RBF architecture consists of the following layers:

- **Input Layer:** Accepts four features (*Latitude*, *Longitude*, *Day of the Week*, *Hour*).
- **RBF Layer:** A custom layer with 10 neurons and Gaussian activation (controlled by a γ parameter of 0.1) to model localized spatial patterns.
- **Output Layer:** A dense layer with softmax activation, outputting the probability distribution over the target classes (crime types).

2) *deep NN-based:* **ConvLSTM-based Architecture:** The architecture processes a 5D tensor (batch, 7, 41, 34, 3) as input and outputs a single future frame (batch, 1, 41, 34, 3).

- **Input Layer:**
 - Accepts a 5D tensor (batch, 7, 41, 34, 3), where 7 is the number of timesteps, and each frame has a spatial dimension of 41×34 with 3 channels.
- **Encoder (ConvLSTM Stacks):**
 - *ConvLSTM2D* (64 filters, 3×3 kernel)

- * Activation: ReLU, padding = same.
- * `return_sequences=True`, so the layer outputs the entire sequence.
- * Followed by BatchNormalization and Dropout(0.3) for stabilization and regularization.

– *Second ConvLSTM2D* (64 filters, 3×3 kernel)

- * Same configuration, with `return_sequences=True`.
- * BatchNormalization and Dropout(0.3).

– *Third ConvLSTM2D* (32 filters, 3×3 kernel)

- * `return_sequences=False`, so only the last time step is returned.
- * BatchNormalization and Dropout(0.3).

– **Flatten**

- * Converts the spatial dimensions (e.g., $41 \times 34 \times 32$) into a single 1D latent vector.

- **Fully Connected Layers (Decoder):**

– *Dense*(128, ReLU) + Dropout(0.3)

- * Compresses the flattened vector into 128 units.

– *Dense*($41 \times 34 \times 3$, Sigmoid)

- * Generates the output in a flattened form for a single future frame.

– **Reshape** $\rightarrow (1, 41, 34, 3)$

- * Reshapes the output vector into one frame with the original spatial dimensions and channels.

- **Output Layer:**

- Produces a single future frame of size (batch, 1, 41, 34, 3).

D. Training and Testing Process

1) *RBF:* For the training process, a sliding time window of 4 weeks was used. The window moved forward by one week after training the model, and the following week was predicted. Then, the model was retrained, considering the new position of the next week to predict the upcoming week. This process was repeated a total of 4 times, meaning the model was trained 4 times and 4 weeks were predicted.

- **Data Split:** The dataset was split into 80% training and 20% testing subsets using a stratified approach to ensure balanced class distribution.
- **Optimization:** The model was compiled with the Adam optimizer (learning rate = 0.001) and *sparse categorical crossentropy* loss, appropriate for multi-class classification tasks.
- **Early Stopping:** Early stopping was implemented with a patience of 5 epochs, monitoring the validation loss to prevent overfitting.
- **Batch Size and Epochs:** The model was trained with a batch size of 32 over a maximum of 10 epochs.

2) *ConvLSTM:* For the training process, a sliding time window of 2 days was used to predict the subsequent day. The input consists of “video-like” data representing daily crime counts in a 1×1 km grid across the city of Chicago, sized

41×34 pixels with 3 channels. The dataset spans 7 years, and for this particular experiment, all of 2024 was used to train the model in order to predict crime occurrences in January 2025. The key steps and parameters are summarized below:

- **Data Representation:** Each day is mapped onto a 41×34 grid (1×1 km cells), with 3 channels representing different crime-related features.
- **Sliding Window (7 days → next day):** A window of 2 consecutive days serves as input frames (timesteps) for the ConvLSTM, and the model predicts the next day (the subsequent frame). After each prediction, the window shifts by one day and the model is trained again if necessary.
- **Training and Testing Split:**
 - The dataset was split into 80% for training and 20% for testing.
 - A 1-month validation set was used to monitor performance and avoid overfitting.
 - In the final stage, the entire year 2024 was used for training to forecast January 2025.

E. Performance Evaluation Metrics

The performance of the model was evaluated using:

- **Accuracy:** The primary metric to evaluate the classification performance.
- **Loss:** Mean cross-entropy loss was monitored to assess model convergence.
- **Training/Validation Curves:** Loss and accuracy curves were analyzed to evaluate model performance over epochs.

VII. PRELIMINARY RESULTS AND PERFORMANCE EVALUATION

A. RBF Results

1) Observations from Training and Validation Performance:

The training and validation performance across four weeks are summarized as follows:

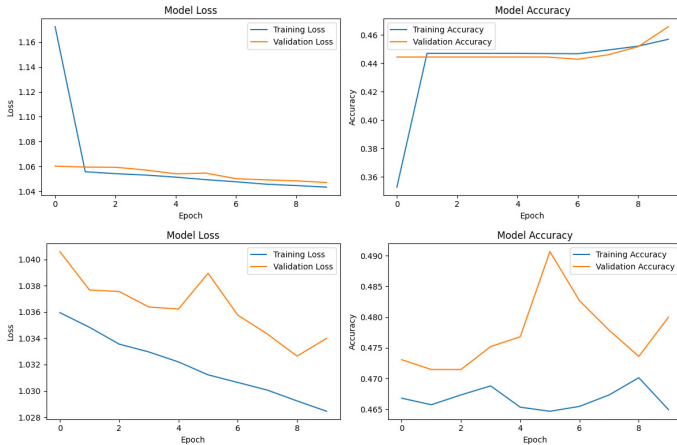


Fig. 2. Model Loss and Accuracy for Week 1 (upper) and Week 2 (lower).

2) *Week 1:* The model loss for both training and validation decreases steadily, indicating an improvement in the fit and generalization of the model to the data. This suggests that the network was able to learn meaningful patterns without overfitting during this period (fig. 2).

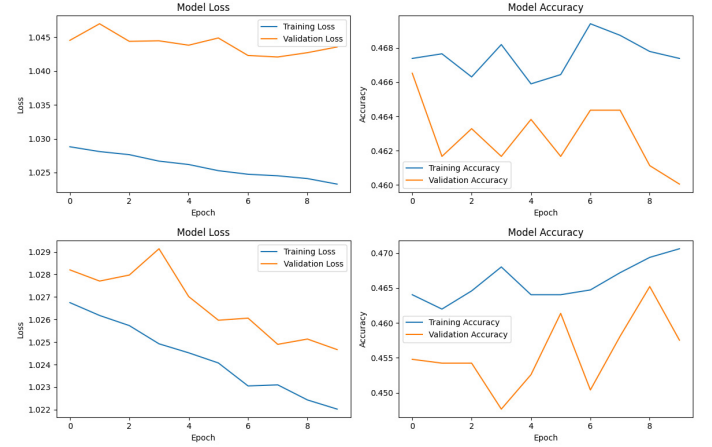


Fig. 3. Model Loss and Accuracy for Week 3 (upper) and Week 4 (lower).

3) *Week 2 and Week 3:* In these weeks, a notable divergence between training and validation loss is observed:

- The training loss continues to decrease, showing the model's ability to optimize for the training data.
- The validation loss, however, remains stagnant or even increases, suggesting that the model begins to overfit to the training data.

This overfitting behavior is further corroborated by the validation accuracy, which does not improve and remains significantly below 50%, (fig. 2, fig. 3).

4) *Week 4:* The trend of overfitting becomes more pronounced in Week 4:

- The training loss decreases consistently, while the validation loss remains almost constant, indicating a strong overfitting to the training data.
- The validation accuracy continues to stagnate, demonstrating the model's inability to generalize to unseen data effectively.

The validation accuracy does not surpass 50%, highlighting the poor performance of the current architecture in modeling crime occurrences (fig. 3).

5) *Diagnosis of Model Issues:* From the observed trends in loss and accuracy, the following issues are evident:

- **Overfitting:** The model shows clear signs of overfitting, particularly in Weeks 2, 3, and 4. This is evidenced by the growing gap between training and validation loss.
- **Low Accuracy:** The validation accuracy remains below 50%, which is inadequate for a crime prediction task. This suggests that the current neural network architecture is not suitable for modeling the complex relationships in the data.
- **Generalization Failure:** The model fails to generalize to unseen data, as indicated by the stagnant or increasing

validation loss and the lack of improvement in validation accuracy.

The current shallow neural network architecture demonstrates significant limitations in its ability to model the complexity of crime prediction tasks. The observed overfitting and low accuracy highlight the need for substantial adjustments to the model design and training strategy. Future iterations should prioritize improved regularization, simplified architectures, and alternative methodologies to address these challenges effectively.

B. ConvLSTM Results

The model's loss function steadily decreases for both training and validation, converging around 0.046. Notably, the training and validation curves follow a similar pattern, suggesting that the model is generalizing effectively, as illustrated in Figure 4. Regarding accuracy, standard classification metrics are not entirely suitable for this task, given that the model generates an image and compares it to the ground-truth crime distribution for a single day. However, the classification-based accuracy metric—used here simply as a reference—reaches approximately 0.9 for both training and validation, indicating strong performance despite the limitations of this metric for image-generation tasks.

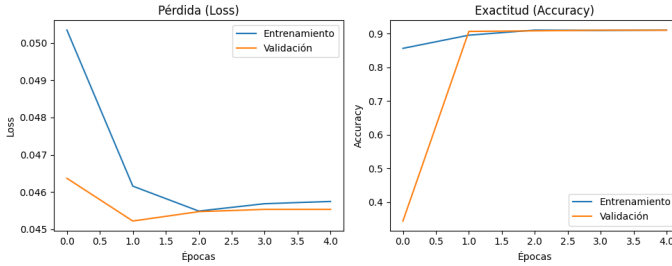


Fig. 4. Model Loss and Accuracy for ConvLSTM.

VIII. ETHICAL CONSIDERATIONS/QUESTIONS

The development of predictive models for crime activity raises important ethical questions and considerations. These concerns are crucial to ensure that the solutions proposed do not perpetuate or exacerbate existing inequalities or biases. Below, we discuss key aspects of ethical considerations with a focus on justice between populations.

A. RBF implications

1) *Observations on Justice Between Populations:* The results obtained from the model highlight significant disparities in predictive performance between protected and unprotected populations for the analyzed crimes:

- **Battery (Battery Criminal Activity):** The prediction error is consistently higher for unprotected populations, as shown in Figure fig. 5.
- **Criminal Damage (Criminal Damage Crime):** Similar to battery, the prediction error is disproportionately higher for unprotected populations, as shown in Figure fig. 6.

- **Theft (Theft Crime):** The model demonstrates a clear bias, with higher prediction errors for unprotected populations, as shown in Figure fig. 7.

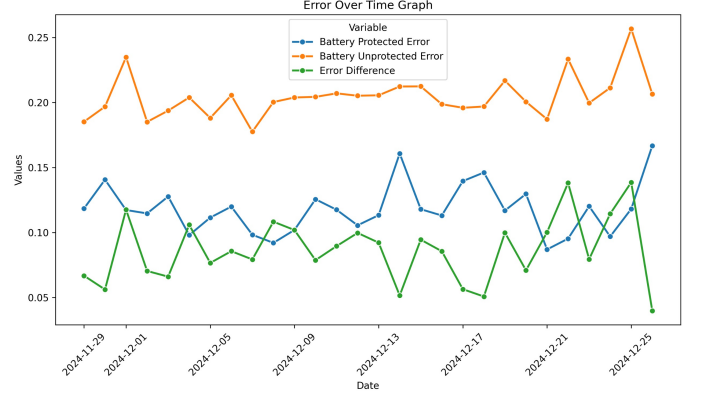


Fig. 5. Prediction Errors for Battery Criminal Activity (Protected vs. Unprotected Populations).

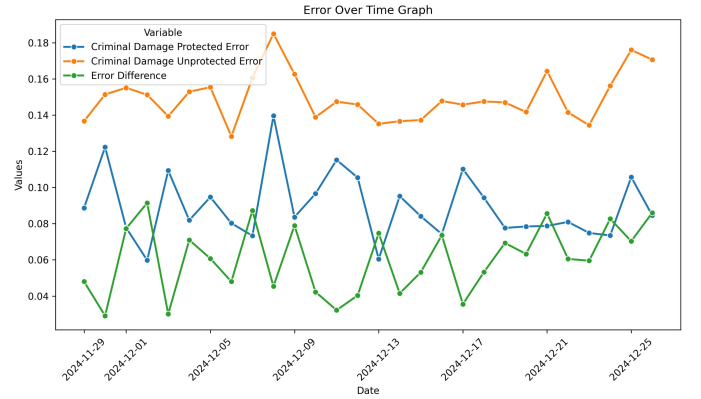


Fig. 6. Prediction Errors for Criminal Damage Crime (Protected vs. Unprotected Populations).

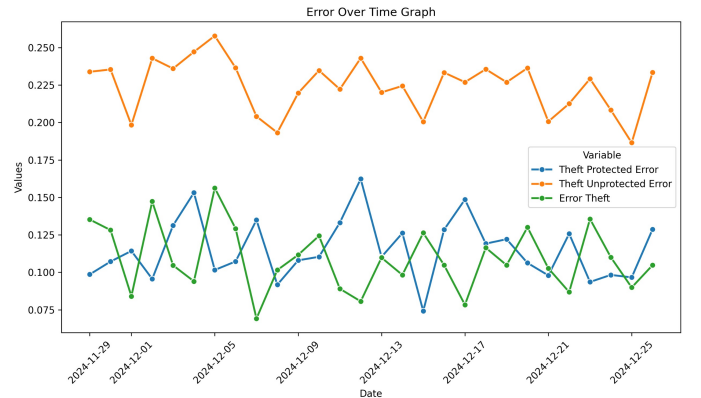


Fig. 7. Prediction Errors for Theft Crime (Protected vs. Unprotected Populations).

These trends indicate that the model systematically underperforms for unprotected populations, raising concerns about fairness and equity in its application.

B. ConvLSTM implications

1) *Observations on Justice Between Populations:* The comparison with the RBF model yields mixed results across different crime categories. In some scenarios, the error is higher in protected areas than in unprotected ones, while in other cases, the opposite is observed.

- **Battery (Battery Criminal Activity):** The prediction error is consistently higher for protected populations, as shown in Figure fig. 8.
- **Criminal Damage (Criminal Damage Crime):** Similar to battery, the prediction error is disproportionately higher for protected populations, as shown in Figure fig. 9.
- **Theft (Theft Crime):** The model demonstrates a clear bias, with higher prediction errors for unprotected populations, as shown in Figure fig. 10.

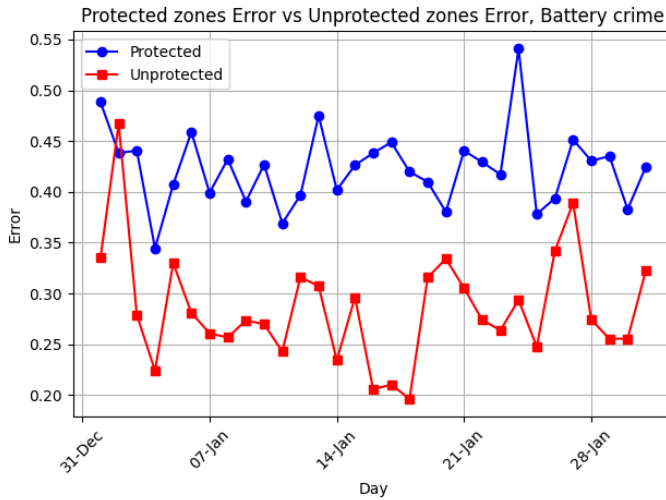


Fig. 8. Prediction Errors for Battery Criminal Activity (Protected vs. Unprotected Populations).

These observations suggest that the model systematically underperforms for unprotected populations, raising significant concerns about fairness and equity in its practical application.

C. Potential Causes of Disparities

The observed disparities may be attributed to multiple factors:

- **Resource Allocation:** In the United States, areas with higher tax revenues tend to have better law enforcement resources, including more frequent patrolling. This imbalance likely contributes to the model's poorer performance in underprotected areas.
- **Bias in Data Reporting:** Crime data may be biased due to underreporting in unprotected populations. According to Adolphe Quetelet's theory of the "average man," environmental factors heavily influence average

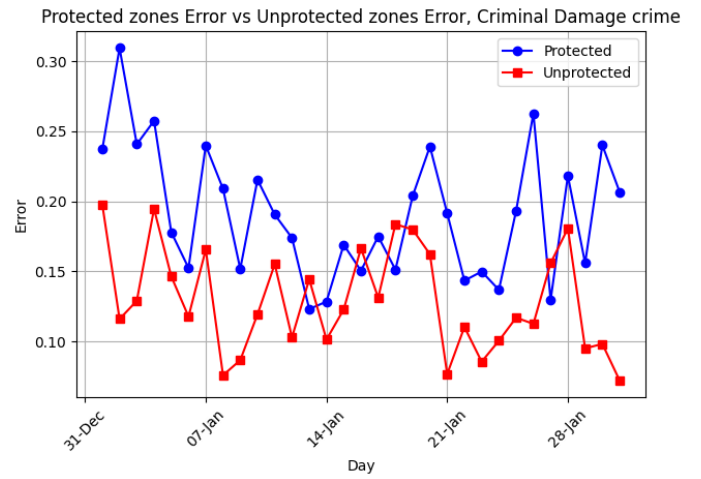


Fig. 9. Prediction Errors for Criminal Damage Crime (Protected vs. Unprotected Populations).

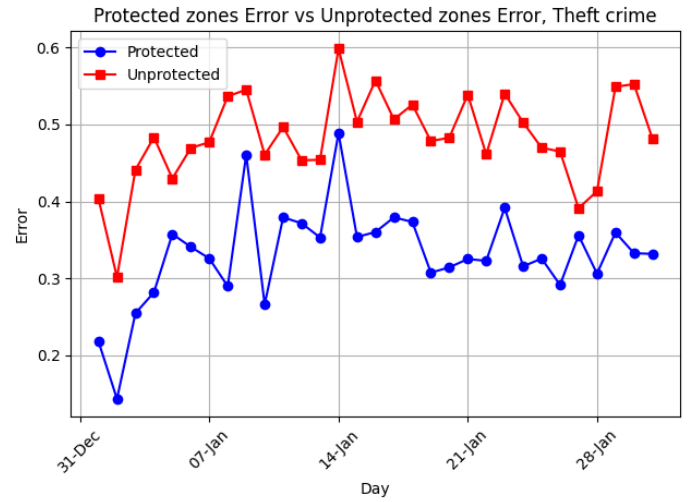


Fig. 10. Prediction Errors for Theft Crime (Protected vs. Unprotected Populations).

actions. In this context, certain crimes may be deemed less important by communities in underprotected areas and thus go unreported, further exacerbating data bias.

D. Ethical Questions to Consider

The following ethical questions should guide the development and application of crime prediction models:

- **Voluntary Participation and Consent:** Are the communities affected by these predictions informed and consenting participants in the process? How can their input be integrated into model design and deployment?
- **Anonymity and Confidentiality:** Does the use of geographic and demographic data respect the anonymity and confidentiality of the individuals and communities involved?

- **Harm and Equity:** Does the model perpetuate systemic inequalities by systematically disadvantaging underprotected populations? How can these harms be mitigated?
- **Transparency and Communication:** Are the results and limitations of the model communicated transparently to stakeholders, including law enforcement agencies and affected communities?
- **Bias Mitigation:** What measures can be implemented to reduce bias in the data and the model? For example, could reweighting techniques or fairness-aware algorithms be applied?

E. Ethical Research Practices

To address the ethical concerns highlighted above, the following recommendations are proposed:

- Engage with communities to understand their perspectives and ensure that the solutions address their specific needs.
- Incorporate fairness-aware algorithms and metrics to reduce disparities in model performance between populations.
- Increase transparency by openly communicating the limitations of the model, especially in terms of bias and accuracy.
- Regularly audit the model to ensure that it aligns with ethical principles and does not inadvertently harm vulnerable populations.

IX. WEB LINKS AND RESOURCES

The project's resources will be made available through the following channels:

- Project Repository: [Grupo-5-Fairness-in-Crime GitHub](#)
- Video Presentation: [Grupo-5-Fairness-in-Crime Youtube](#)

X. CONCLUSION

The analysis reveals significant disparities in prediction errors between protected and unprotected populations for battery, criminal damage, and theft. Unprotected populations consistently experience higher errors, likely due to systemic issues such as uneven resource allocation and biases in crime reporting.

When comparing models, the ConvLSTM approach—while less prone to overfitting than the RBF model—exhibited overall higher prediction errors. Notably, the RBF model underperforms for unprotected populations, whereas the ConvLSTM model shows higher errors for protected populations in battery and criminal damage, but higher errors for unprotected populations in theft.

These findings raise ethical concerns, as predictive models that disproportionately affect vulnerable communities may reinforce existing inequalities. Addressing these issues through fairness-aware algorithms, robust auditing, and increased community involvement is essential to develop predictive systems that are both accurate and equitable.

ACKNOWLEDGMENTS

Tools Used: Claude, ChatGPT, and QuillBot were part of the process for drafting, writing, and editing this document. They made everything much easier and smoother!

A. Contribution of the team members

All team members made equal contributions to the development of this project proposal as we can see in table III. This idea originates from the thesis proposal development of PhD student Diego Alejandro Hernández Castañeda.

Team Member	Role	Contributions
Diego A. Hernández C.	Lead	Propose coordinate idea Create document Search for information
Jose L. Moreno H.	Expert	Propose coordinate idea Create document Search for information
David F. Ademes R.	Expert	Propose coordinate idea Create document Search for information

TABLE III
TEAM ROLES AND CONTRIBUTIONS

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